TeamSync

an AI assistant for internal task tracking and team visibility.

1. Core Problem

Work is discussed across multiple internal groups and private conversations, but there is no consolidated view of the commitments made in those conversations.

As a result:

- Tasks and initiatives get buried in messages and meeting notes.
- Employees repeatedly explain the same progress to different people.
- Team members have limited visibility into what colleagues are working on.
- Managers must chase updates before they can identify blockers or redistribute work.
- Cross-functional projects lack a shared, current view of responsibilities and progress.

The core problem is the repeated effort required to turn scattered conversations into shared, reliable work status.

2. Product Overview

TeamSync connects internal conversations to shared project trackers and a consolidated team dashboard.

A visible bot proposes tasks and updates from enabled group conversations. In private chats, users explicitly select the messages they want processed.

Approved tasks appear in the relevant group or private-chat tracker and in the responsible team's Orbyte dashboard, according to permissions.

One task, multiple authorized views: Updating a task updates every place it appears. These are not separate copies.

3. Goals and Boundaries

Goals

- Reduce repetitive status reporting and manual task entry.
- Give teams and managers an accessible view of ongoing work.
- Track cross-functional commitments without exposing entire conversations.
- Preserve human control over AI access, task changes, and sharing.

Boundaries

- Internal collaboration only; external communication integrations are excluded.
- The assistant prepares suggestions; authorized people approve AI-generated changes.
- Visibility into tasks does not grant visibility into their source chats.
- Task counts indicate commitments, not actual capacity. Reliable workload balancing also requires information about effort, availability, and complexity.

4. Primary User Journey

Meet the team: Emma, James, and Sarah work on a team managed by David. Each participates in different project groups, including groups with other teams.

The journey follows how work moves from **conversation → confirmed task → shared visibility → management oversight**.

Step 1: Discuss work where it happens

The team collaborates with another department in a shared project group. After a meeting, Sarah posts the notes and agreed next steps.

A group admin enables TeamSync. Everyone sees that the assistant is active. It analyzes new messages; earlier notes can be submitted explicitly.

Step 2: Turn the discussion into confirmed tasks

TeamSync identifies proposed actions, owners, and deadlines. The designated project editors review the suggestions, resolve missing details, and approve them.

The approved tasks appear in the group’s tracker and the responsible team’s dashboard.

Step 3: Capture commitments from private conversations

Emma agrees on a follow-up task with Michael from another team in a private chat.

She highlights the relevant messages, selects **“Jot this down”,** and reviews the proposed task and intended audience before sharing it.

Once approved, the task appears in her team’s dashboard and her private-chat task panel. Michael sees it only if explicitly granted access. The rest of their conversation remains private.

Step 4: Catch up without asking everyone for updates

A member returning to the project opens the group’s top bar to see accessible ongoing tasks, pending approvals, and recent changes.

Emma opens her team dashboard to see what James and Sarah are working on—even when their tasks originated in groups she does not belong to. She sees the shared work records, not those conversations.

Step 5: Manage work from a consolidated view

David reviews the team dashboard to understand progress, blockers, deadlines, and overlapping commitments before asking for updates.

Team members update tasks within their permissions. Cross-functional task changes remain controlled by the designated project editors, who can delegate that responsibility.

Step 6: Extend visibility when needed

David grants a senior manager **Viewer** access to the team dashboard. If someone needs to help maintain the work, he can grant scoped **Editor** access.

Everyone works from the same task records, with visibility and editing rights appropriate to their role.

5. Feature Requirements

No.	Feature	Requirement	Permissions and Controls
1	Visible Group Bot	Allow a group admin to add, enable, pause, or remove TeamSync. Display a notice and persistent indicator when AI analysis is active.	Analyze new messages only while enabled. Earlier messages require explicit submission.
2	AI Task Suggestions	Identify proposed tasks, owners, deadlines, and updates from group messages and meeting notes.	Require clarification for missing or ambiguous information. Human approval is required before applying changes.
3	Private-Chat Capture	Let users highlight messages and select “Jot this down.” Preview the extracted proposal and intended audience before sharing.	Process only selected messages. Do not monitor private chats or share original excerpts with reviewers.
4	Group Project Tracker	Open Overview, Tasks, Pending Approvals, and Recent Updates from the group’s top bar.	Show only records the viewer can access, including counts and activity previews.
5	Private-Chat Task Panel	Display tasks linked to a private conversation in its task panel.	Each participant sees only tasks explicitly accessible to them. Chat participation does not automatically grant task access.
6	Consolidated Team Dashboard	Combine the team’s tasks across projects, groups, and private-chat submissions. Show owners, statuses, deadlines, and blockers.	Team task visibility does not grant access to source conversations.
7	Task Status Updates	Support Not started, In progress, Blocked, On hold/KIV, and Done.	Team members may update team-only task statuses. Cross-functional tasks require designated project editors.
8	Approval Workflow	Allow authorized reviewers to approve, edit, reject, or request clarification on AI proposals. Show current and proposed values.	Only designated approvers can apply changes. Outdated proposals require renewed review.

9	Consistent Task Records	Maintain one task record across group trackers, private-chat panels, and team dashboards.	Updates appear consistently across authorized views. Prevent duplicate creation or execution.
10	Cross-Functional Coordination	Track shared project commitments across participating teams.	Designated project managers and delegated editors control changes across every view. Others may propose updates.
11	Sharing and Delegation	Grant Viewer or Editor access to specific tasks, projects, or team dashboards.	Access must be explicit, scoped, revocable, and logged. Task sharing does not include source-chat access.
12	Activity History	Record task changes, approvals, editors, timestamps, and previous values.	Display only authorized activity and exclude restricted source messages.
13	Catch-Up Context	Provide a group purpose, maintained project overview, and recent approved updates for returning members and new joiners.	Do not summarize inaccessible conversations or automatically expand access when someone joins.
14	Failure Handling	Clearly show failed analysis or task updates and provide a retry option.	Preserve submitted information, prevent duplicate updates, and never display a failed action as successful.

6. Permissions and Editing Rules

Area	**Who Can View?**	**Who Can Edit?**
Team Dashboard	Team members, reporting manager, and explicitly invited viewers.	Team members can update team-only task statuses. Managers and delegated editors can change ownership, deadlines, and scope.
Cross-Functional Project Tasks	Authorized participating teams and explicitly invited users.	Designated project managers and delegated project editors. Other members can propose updates for approval.
AI Proposals	The submitter and authorized destination reviewers.	Submitters can revise drafts. Designated approvers can approve, edit, or reject proposals.

Private-Chat Source Messages	Existing participants in the private conversation.	Governed by chat permissions. Sharing a task does not grant access to its source messages.
Shared Management View	Managers or other users explicitly invited to the relevant dashboard or project.	Viewers have read-only access. Explicitly assigned Editors can modify records within their granted scope.

Important consistency rule: Cross-functional task restrictions apply everywhere. A team member cannot bypass them by editing the same task from the team dashboard.

Members without editing rights can submit a proposed progress update for an authorized editor to apply.

7. Reliability Requirements

- Distinguish an agreed action from a question, suggestion, or hypothetical discussion.
- Never invent an owner, deadline, or completion state.
- Prevent duplicate tasks from repeated processing or approval.
- Require renewed review when underlying messages or task records change.
- Show processing and update failures clearly.
- Enforce access on the backend, including tracker counts and activity previews.
- Do not disclose restricted task details in group bot replies.
- Preserve an audit trail for approved changes and permission grants.

8. Technical Approach and Delivery

Technical approach. TeamSync is a React + TypeScript web app backed by a single FastAPI service (REST and WebSockets) and PostgreSQL as the only datastore. Every change writes an event row in the same transaction, and an AI worker reads that event log, debounces by channel, assembles context, and calls the model through a tool registry. Each tool has a JSON schema and a required scope. The dispatcher validates arguments, checks scopes, and logs every call. The AI only proposes tasks, and a person accepts them.

9. Success Criteria and Open Risks

Acceptance criteria

- An approved task appears consistently in the appropriate trackers and team dashboard.
- Users can find current work without searching message history.
- Managers can inspect status before requesting an update.
- Unauthorized users cannot view or modify restricted records.
- Private-chat capture shares only the reviewed proposal.
- Cross-functional editing restrictions hold across every view.
- No AI-generated change executes without approval.

Validation of user workflow

Measure time spent capturing tasks, correcting suggestions, and reporting status against the existing process.

The main unresolved risks are whether employees will move internal discussions into Orbyte, whether they keep statuses current, and whether manager approval becomes a bottleneck. The tool can reduce repeated reporting only when its task records remain trustworthy.